

Customer Purchase Prediction Report

Zeravia — Machine Learning with Python | Mini Project Report

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1. What This Project Is About

The client is a retail business currently sending the same marketing offers to every customer, regardless of whether they're actually likely to buy. That's inefficient — a lot of budget goes toward people who were never going to convert anyway.

Goal: Build a model that predicts whether a given customer is likely to purchase (Purchased = Yes/No), so the client can target campaigns smarter instead of blasting everyone equally.

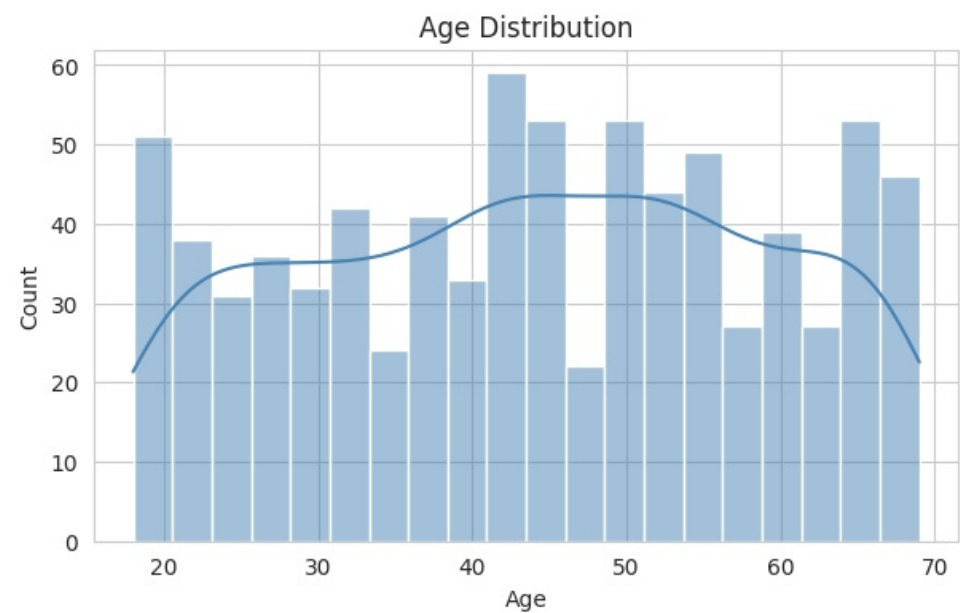
2. About the Dataset

Since no real client dataset was handed over for this assignment, I generated a synthetic dataset (retail_customer_data.csv, 800 records) built to realistically resemble what a retail company would actually track: demographics (Age, Gender, Region), engagement (Engagement_Score, Avg_Session_Minutes), and purchase history (Purchase_Frequency, Discount_Usage_Rate, Total_Spend), with Purchased as the target column. I deliberately built in real-world messiness — missing values, duplicate rows, inconsistent labels — so the cleaning step below is genuine, not just for show.

3. Cleaning the Data

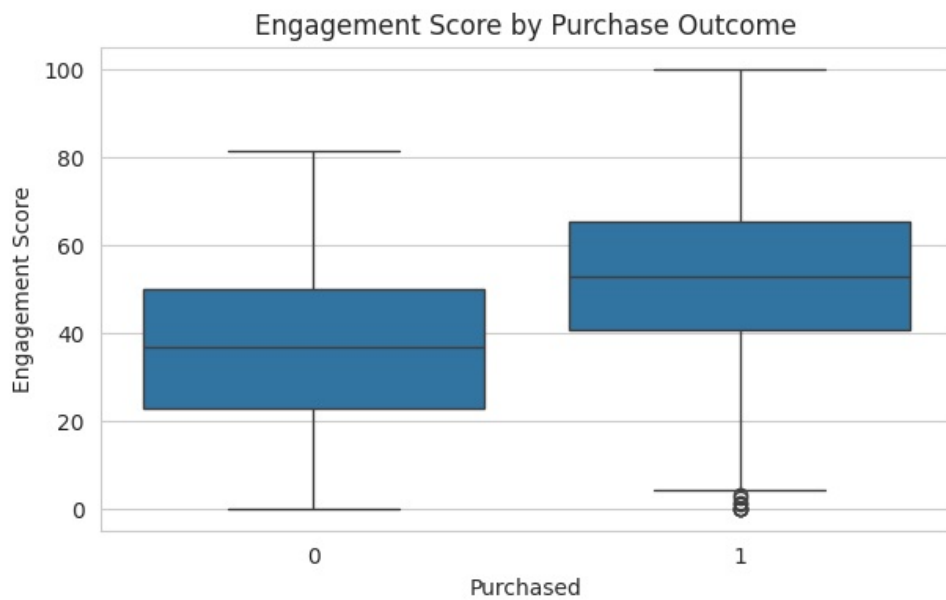
- Dropped duplicate customer rows.
- Standardised inconsistent Gender entries (e.g. "M", "male" → "Male").
- Fixed sign-flipped Age values (clearly a data-entry error, not a real value) and treated invalid negative Annual_Income as missing instead of guessing.
- Filled missing numeric values with the median (less sensitive to outliers than the mean).
- Engineered a few extra features: Age_Group, Spend_Band, Engagement_Level, Frequent_Buyer.
- Scaled numeric features with StandardScaler and one-hot encoded the categorical columns before modelling.

4. What the Data Shows



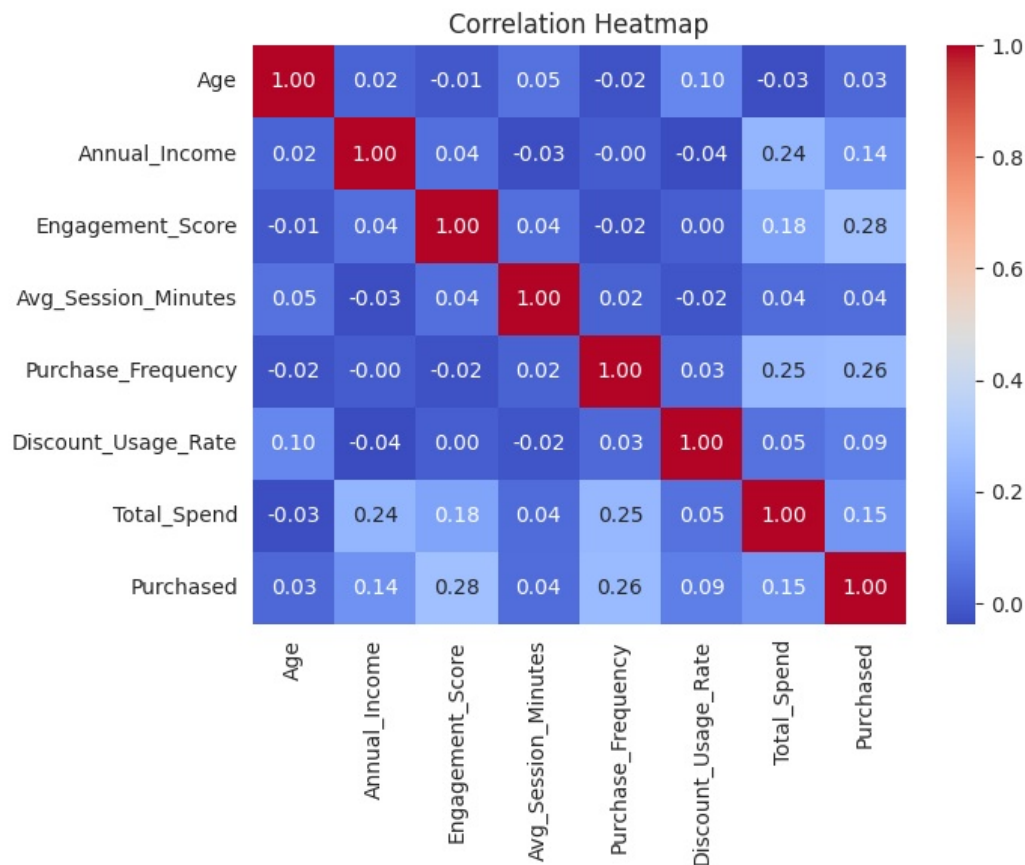
Age Distribution

Ages are spread out pretty evenly across 18-70, no real spike anywhere — so age on its own probably isn't a strong predictor of purchase.



Engagement Score by Purchase Outcome

Customers who purchased have noticeably higher engagement scores than those who didn't. This is a good sign that engagement carries real predictive signal.



Correlation Heatmap

Purchase_Frequency and Engagement_Score show the strongest relationship with Purchased, while Age barely correlates at all — matches what the age chart above already suggested.

Income relates to spend, but doesn't cleanly separate purchasers from non-purchasers by itself.



Purchase Rate by Region

Purchase rate isn't the same across regions — some regions convert noticeably better than others, so a single blanket campaign strategy probably isn't ideal.

5. Stats & Outliers

Mean, median, and standard deviation were checked for the key numeric columns. Using the IQR method, a handful of outliers turned up in `Total_Spend` and `Annual_Income` — these look like genuine high-spending customers rather than data errors, so I kept them in the dataset but scaled the features before modelling to reduce how much they could skew things.

6. Building the Models

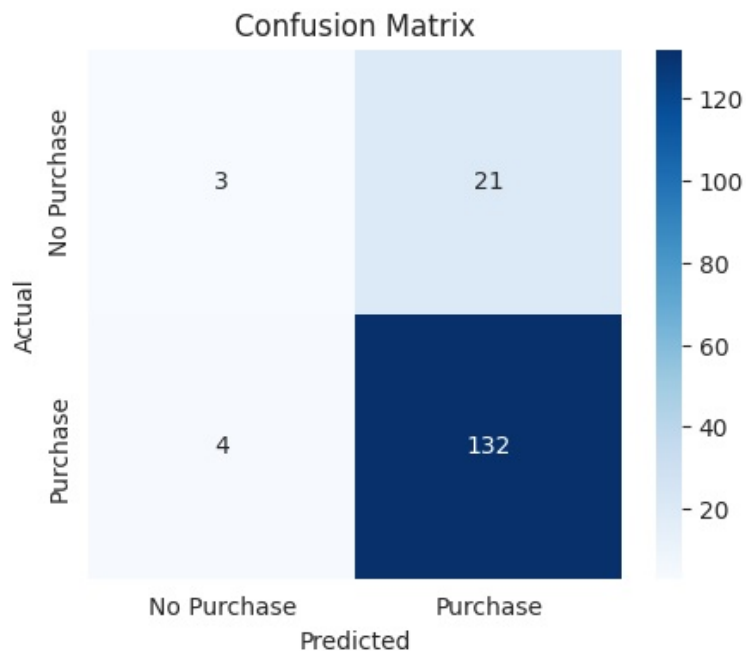
Problem type: Binary classification, supervised learning.

Since `Purchased` is a 0/1 column, Linear Regression isn't really the right tool for it — it's built for continuous numbers, not yes/no outcomes. So Linear Regression was applied instead to `Total_Spend`, which is a genuine continuous target in the data:

Metric	Value
MAE	638.34
MSE	657,906.35
R ²	0.421

The actual model answering the client's question is Logistic Regression, predicting `Purchased`:

Metric	Value
Accuracy	0.844
Precision	0.863
Recall	0.971
F1-score	0.913



Confusion Matrix

What this means: Precision tells us, of the customers predicted to buy, how many actually did. Recall tells us how many of the actual buyers were caught by the model. There's a trade-off between the two, and the confusion matrix shows exactly where the model gets things right or wrong. The model isn't perfect — Logistic Regression draws a fairly simple decision boundary, so it may be missing more complex patterns in customer behaviour, and this is on synthetic data rather than real client records, so results would likely shift with real data and more tuning.

7. What I'd Recommend to the Client

1. Focus on customers with high engagement scores — they're most likely to buy, based on the analysis.
2. Give loyalty offers to customers who already buy often, since past purchases predict future ones fairly well.
3. Purchase rates differ across regions, so maybe don't run the exact same campaign everywhere.
4. Give small discounts to Low/Medium spenders to encourage them, but Very High spenders probably don't need discounts to keep buying.
5. Customers who use discounts more also tend to purchase more, so discount codes could help convert medium-engagement customers.
6. Instead of a plain yes/no, the model's probability score could be used to rank and prioritize customers.
7. Update the model every few months since customer behaviour changes over time.

8. Limitations

No real client dataset was available for this project, so a synthetic dataset was generated to realistically resemble retail customer data, including genuine data-quality issues so the cleaning workflow wasn't just for show. Tree-based models and neural networks were intentionally left out, since they're outside the current curriculum scope (scheduled for later months). The numbers here are meant to show the workflow rather than represent a production-ready model.